

C Programming Practice Sheet (Week -1)

Q1. Student Information

Write a C program to input and display a student's **name, age, roll number, and percentage**.

Q2. Add Two Numbers

Write a C program to input two integers and display their **sum**.

Q3. Personal Details

Write a C program to input your **name, age, city, and college name** and display them in a formatted manner.

Q4. Calculate Square

Write a C program to input a number and calculate and display its **square**.

Q5. Calculate Cube

Write a C program to input a number and calculate and display its **cube**.

Q6. Area of Rectangle

Write a C program to input the **length and breadth** of a rectangle and calculate its **area** and **perimeter**.

Q7. Area of Circle

Write a C program to input the **radius** of a circle and calculate its **area and circumference**.

Use:

- $\text{Area} = \pi \times r \times r$
- $\text{Circumference} = 2 \times \pi \times r$

Q8. Temperature Conversion

Write a C program to input temperature in **Celsius** and convert it into **Fahrenheit**.

Formula:

$$F = (C \times 9/5) + 32$$

Q9. Simple Interest

Write a C program to input **Principal, Rate, and Time** and calculate Simple Interest.

Formula:

$$SI = (P \times R \times T) / 100$$

Q10. Average of Three Numbers

Write a C program to input three numbers and calculate their **sum and average**.

Q11. Basic Arithmetic Operations

Write a C program to input two numbers and display their:

- Addition
- Subtraction
- Multiplication
- Division
- Remainder

Q12. Arithmetic Calculator

Write a C program that takes two numbers as input and performs **addition, subtraction, multiplication, and division**.

Example:

Enter first number: 20

Enter second number: 5

Result:

Addition = 25

Subtraction = 15

Multiplication = 100

Division = 4

Q13. Calculator Using Operators

Write a C program to input two integers and display the result of:

a + b

a - b

a * b

a / b

a % b

Q14. Total and Percentage

Write a C program to input marks of **5 subjects** and calculate:

- Total marks
- Average
- Percentage

Assume each subject is out of 100.

Q15. Employee Salary Calculation

Write a C program to input **basic salary** and calculate:

- HRA = 20% of basic salary
- DA = 10% of basic salary
- Gross Salary = Basic + HRA + DA

Q16. Shopping Bill

Write a C program to input the **price and quantity of three products** and calculate the total bill.

Example:

Product 1 Price: 100

Product 1 Quantity: 2

Product 2 Price: 50

Product 2 Quantity: 3

Product 3 Price: 200

Product 3 Quantity: 1

Total Bill = ?

Q17. Currency Conversion

Write a C program to input an amount in **Indian Rupees** and convert it into **US Dollars**.

Assume: 1 USD = ₹85

Q18. Time Conversion

Write a C program to input a number of **seconds** and convert it into:

Hours

Minutes

Seconds

Example:

Input: 3665

Q19. Calculator Using Menu

Write a C program that displays the following menu:

===== CALCULATOR =====

1. Addition

2. Subtraction

3. Multiplication

4. Division

5. Modulus

Take the user's choice and two numbers as input, then display the appropriate result.

Q20. Mini Arithmetic Calculator

Write a C program that accepts **two numbers and an operator** (+, -, *, /, %) from the user and performs the selected operation.

Example:

Enter first number: 25

Enter operator: *

Enter second number: 4